# ISA5810 DM2026 Lab 1 — Master Notebook Questions

Fill in this file as you work through DM2026-Lab1-Master.ipynb. Submit this completed file converted to PDF together with the rest of your Lab 1 work.

**Please do not change the style of this document.**

**Question numbers 1–10 follow the same order as the notebook** (top to bottom).

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**Student Information**

Please complete the fields below. For **English Name** and **Chinese Name**, you may fill in **only one** if you do not have the other.


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## Instructions

These questions go with the Master Jupyter notebook (DM2026-Lab1-Master.ipynb). Each question is tied to a specific notebook section. Open the notebook, find **Question N (take home)**, and write your answer here.

We **strongly** suggest running the notebook to learn the lab, but the TAs will **not** audit whether you executed every cell to answer the questions in this document.

**How to answer each question**

1. Find the **notebook section** listed under the question (use the Table of Contents if needed).
2. Read the task; change parameters when asked (e.g. stop_words, thresholds, k, minSup).
3. Write a clear answer in the **Answer** space below each question.
4. Leave the question text unchanged; write only in the answer and disclosure fields.

**Sources, honesty, and AI use**

**Acceptable vs. Unacceptable AI Usage:** Generative AI should serve as an interactive tutor to guide your understanding, not as an automated solver completing the assignment for you.

- **Permitted (AI as Guide & Editor):**
  - Asking AI to explain unfamiliar theoretical concepts, discuss algorithmic trade-offs, or troubleshoot runtime errors in the notebook.
  - Polishing the grammar, phrasing, and readability of your own original, self-written drafts.

- **Prohibited (AI as Solver):**
  - Copying assignment questions directly into an AI tool to generate answers.
  - Using AI to fabricate experimental results or observations without executing the code in DM2026-Lab1-Master.ipynb.
  - Submitting machine-generated prose wholesale.

- **External material:** If you use websites, books, papers, or documentation beyond the notebook, cite them with **clickable links** (or DOI/ISBN when no URL exists).

- **When AI was used:** Your answer is acceptable if you (1) **disclose** AI use and (2) include **at least one external link** that validates the information in your answer (official docs, textbook, paper, or reputable tutorial — not the AI chat alone as the only source).

- **When AI was not used:** You may write "No" under **AI used** or leave it blank only if you truly did not use AI for that question. References are still encouraged if you looked anything up.

Under each question, complete:

- **AI used for this answer:** Yes / No
- **If yes, how:** (e.g. brainstorming, grammar, concept explanation)
- **External references (links):** *(required when AI = Yes; encouraged otherwise)*

Write in English. Keep answers concise and specific.

## Questions

**Question 1: Data Filtering & Domain Relevance**

**Notebook:** Section 3.2 — Question 1 (take home)

**Task:** Filter the dataframe **X** for the category **sci.med** and check its length.

**Questions:**

1. How many documents are in this specific category?
2. From a data mining perspective, why is isolating specific categories a crucial first step before applying algorithms like frequent pattern mining?

**Answer:**

1. 594 documents

2. Isolating the categories can make us easier to indentify the pattern since we can control the properties to be the same. Then, we can better view the main property's pattern.

**AI used for this answer:** No

**If yes, how:**

**External references (links):**

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**Question 2: Handling Missing Data**

**Notebook:** Section 4.1 — Question 2 (take home) (after the **dropna()** demonstration)

**Task:** In Section 4.1, a dummy record with a missing value is injected and then removed using the **dropna()** function.

**Question:** If 40% of documents were missing the **category_name** label, why would simply dropping these rows (with **dropna()**) be a flawed data mining strategy? Name one objective alternative so the text data can still be utilized.

**Answer:** Dropping the rows could cause problems. For example, the document might contain categories that aren't mentioned in the remaining documents, so we could miss an entire category. Another problem is that the data with complete label could present only partial pattern of the category. An alternating method (for the second problem) would be estimating missing values (as mentione in the jupyter notebook). We can find a pattern from a category using the remaining 60% of documents first. Then, from the documents with missing value, add the ones that look similar to the pattern back to the dataset. Finally, apply data mining on the mixing data, and compare the two patterns.

**AI used for this answer:** No

**If yes, how:**

**External references (links):**

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**Question 3: Data Sampling Risks**

**Notebook:** Section 5.1 — Question 3 (take home)

**Task:** In Section 5.1, the code uses **X_sample = X.sample(n=1000)** to create a subset. Compare the bar chart of the original category distribution to the sampled distribution.

**Questions:**

1. Do the exact category proportions perfectly match the original?
2. What is the specific risk of simple random sampling on a highly imbalanced dataset?

**Answer:**

1. No, though the proportion look close on the plot, they are not the same.

2. Since simple random sampling chose the sample randomly, the class with higher proportion in the dataset is more likely to be chosen, causing the result to be tended to the pattern of the class with higher proportion.

**AI used for this answer:** No

**If yes, how:**

**External references (links):**

https://www.geeksforgeeks.org/machine-learning/what-is-imbalanced-dataset/

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**Question 4: Text Vectorization and Matrix Sparsity**

**Notebook:** Section 5.3 — Question 4 (take home)

**Task:** In Section 5.3, the **CountVectorizer** converts text into a document-term matrix using **stop_words='english'**. Change this parameter to **stop_words=None** and re-run the cell to check the shape of **X_counts**.

**Questions:**

1. Does the total number of features (columns) increase or decrease?

2. Since stopwords are highly common words that appear in almost all documents, what effect does adding them back have on the overall sparsity (the percentage of zeros) of the matrix?

**Answer:**

1. Increases

2. The sparsity would decrease, since most documents would have the common words, making non-zero percentage of the matrix to increase.

**AI used for this answer:** No

**If yes, how:**

**External references (links):**

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**Question 5: Feature Selection — Variance Thresholding**

**Notebook:** Section 5.3.1 — Question 5 (take home)

**Task:** In the variance filtering section, re-run **variance_filter** using threshold values **threshold = 0.0, threshold = 0.01, and threshold = 0.05**. Record the number of features marked as **keep == True**.

**Questions:**

1. What happens to the total size of the vocabulary as the variance threshold increases from 0.0 to 0.05?
2. In terms of document frequency, what kinds of words have variance close to zero and are eliminated first?
3. Why is eliminating near-zero variance features considered a safe way to reduce dimensionality without harming model accuracy?

**Answer:**

1. The total size decreases.

2. The words that have low frequency.

3. Words with lows variance have little number difference between documents, so they are less important when considering model accuracy.

| Threshold | Number of terms kept (keep == True) |
|---|---|
| 0.0 | 35482 |
| 0.01 | 5089 |
| 0.05 | 1491 |

**AI used for this answer:** No

**If yes, how:**

**External references (links):**

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**Question 6: Feature Correlation — Linear (Pearson) vs. Monotonic (Spearman)**

**Notebook:** Section 5.3.2 — Question 6 (take home)

**Task:** Run the correlation filter against the binary indicator for the **comp.graphics** class. Inspect the top 10 features sorted by **pearson_r** and compare them to the top 10 features sorted by **spearman_r**.

**Questions:**

1. Are the top-correlated words generic stop words or domain-specific terminology for graphics?
2. What is the mathematical difference in what **Pearson** correlation measures versus what **Spearman** rank correlation measures?
3. In what specific scenario would a word have a high **Spearman** rank correlation with a topic, but a significantly lower **Pearson** correlation?

**Answer:**

1. domain-specific terminology

2. Pearson correlation measures the linear relation while Spearman rank correlation measures monotonic relation

3. If the two data have polynomial relationship.

**AI used for this answer:** Yes

**If yes, how:** Asked about why there are still some words like “people”, “god”, which looks like generic stop words.

**External references (links):**

https://koaning.io/posts/stopwords/

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**Question 7: Frequent Pattern Mining — Algorithm & Parameter Dynamics**

**Notebook:** Section 5.4.2 — Question 7 (take home)

**Task:** In the Frequent Pattern Mining section, run **Top-K FAE** with **k = 100 and k = 500**. Then, run **MaxFPGrowth** using minimum support thresholds **minSup=9 and minSup= 3**.

**Questions:**

1. How does decreasing **minSup** (from 9 to 3) in **MaxFPGrowth** affect the execution runtime and the total number of discovered patterns?
2. How does the parameter k in **Top-K FAE** differ fundamentally from **minSup** in **MaxFPGrowth** when deciding which patterns to keep?
3. Why are "maximal" frequent patterns structurally more concise than standard frequent patterns?

**Answer (1) — Runtime and pattern count when lowering minSup:**

After decreasing the minSup, the execution runtime largely increases and the total number of discovered patterns also largely increases.

**Answer (2) — Top-K k vs MaxFPGrowth minSup:**

The k isn’t a shreshold like minSup, so we can get a certain number of patttern we want. However, this also means that we might get some data with low support might be contained.

**Answer (3) — Maximal vs standard frequent patterns:**

The maximal frequent pattern won’t get choose all the patterns since we can tell that the chosen ones’ subset also have higher frequency.

**AI used for this answer:** Yes

**If yes, how:** Ask about explaination of terms related to this question, like maximal frequent pattern with simple example.

**External references (links):**

https://www.geeksforgeeks.org/dsa/maximal-frequent-itemsets/

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**Question 8: Dimensionality Reduction — Motivation**

**Notebook:** Section 5.5 — Question 8 (take home) (before PCA/t-SNE code)

**Task:** After text vectorization, the **X_counts** matrix creates tens of thousands of distinct features (words).

**Questions:**

1. If we want to visually map how these documents group by category on a scatter plot, why is it mathematically and visually impossible to plot the raw **X_counts** matrix directly?
2. What specific category of data mining algorithms (referenced in the helper files) must be applied first?

**Answer:**

1. X_counts matrix has high dimension, making it hard to plot. Scatter plot are used for two dimentional matrix.

2. Dimensionality Reduction Techniques

**AI used for this answer:** No

**If yes, how:**

**External references (links):**

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**Question 9: Dimensionality Reduction — Linear (PCA) vs. Non-linear Manifolds (t-SNE/UMAP)**

**Notebook:** Section 5.5 — Question 9 (take home) (after 2D/3D plots)

**Task:** In the 2D/3D Dimensionality Reduction section, compare the scatter plot projection generated by **PCA** against that generated by **t-SNE** or **UMAP**.

**Questions:**

1. Which method **(PCA or t-SNE/UMAP**) shows distinct, separated cluster islands for the 4 document categories, and which shows heavy overlap around the center?
2. Why does **PCA** struggle to separate high-dimensional text categories compared to **t-SNE** or **UMAP**?
3. Why must data analysts be careful not to over-interpret the physical distance between separate clusters in a **t-SNE** plot?

**Answer:**

1. t-SNE/UMAP shows distinct, seperated cluster islands, and PCA shows heavy overlap around the center.

2. PCA is good at seperating Guassian-distributed, so this data is probably not Guassian-distributed.

3. Because t-SNE is sensitive to parameters.

**AI used for this answer:** No

**If yes, how:**

**External references (links):**

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**Question 10: One-Hot Binarization**

**Notebook:** Section 5.6 — Question 10 (take home)

**Task:** In Section 3.1, examine **X['category']** (integers 0, 1, 2, 3). Then look at the output of **LabelBinarizer** in Section 5.6, where each category is converted into a binary vector (e.g., [1, 0, 0, 0]).

**Questions:**

1. How many binary columns are created to represent the 4 categories?
2. If an algorithm (like linear regression or KNN) is given the raw integer column **(0, 1, 2, 3)** as input, what false mathematical assumption does it make about the categories?
3. How does one-hot binarization eliminate this false assumption?

**Answer:**

1. 4 columns

2. Categories are qualitative data, they do not have distance or mathmetical interpretation.

3. The categories all have the same distance (though they do not have meaning) between, so categories can be analyzed without weight.

**AI used for this answer:** No

**If yes, how:**

**External references (links):**

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